Exp.No:6
12/9/26

ERROR CORRECTION AT DATA LINK LAYER

Aim:

Write a Program to implement error detection and correction using HAMMING Code concept Make a test run to input data stream and Verify error correction feature

SENDER PROGRAM WITH BELOW FEATURES.

1. Input to sender file should be a text of any length. Program should convert the text to binary.
2. Apply Hamming code concept on the Binary data and add redundant bits to it.
3. Save this output in a filled called channel.

RECEIVER PROGRAM WITH BELOW FEATURES:

1. Receiver program should read the input from the Channel file.
2. Apply hamming code on the binary data to check for Errors
3. If there is an error, display the position of the error.
4. Else, remove the redundant bits and convert the binary data to ASCII data and display the output.

Sender Program (Sender.py)

```
def encode(d):
    c = [0] * 13
    j = 0
    for i in range(1, 13):
        if i not in (1, 2, 4, 8):
            c[i] = d[j]
            j += 1
    for p in (1, 2, 4, 8):
        for i in range(1, 13):
            if i & p and i != p:
                c[p] ^= c[i]
    return ''.join(map(str, c[1:]))

s = input("Enter message: ")
b = ''.join(format(ord(x), '08b') for x in s)
with open("channel", "w") as f:
    for i in range(0, len(b), 8):
        f.write(encode(list(map(int, b[i:i+8]))) + '\n')
print("Data encoded and saved.")
```

RECEIVER PROGRAM

```
def decode(s):
    c = [0] + list(map(int, s))
    e = 0
    for p in (1, 2, 4, 8):
        if sum(c[i] for i in range(1, 13)
if (i & p) != 2:
            e += p
    if e:
        print("Error at position", e)
        c[0] ^= 1
    return ''.join(str(c[i]) for i in range(1, 13)
                   if i not in (1, 2, 4, 8))
with open("channel") as f:
    b = ''.join(decode(s.strip()) for s in f)
print("Received message:", ''.join(chr(int(b[i:i+8], 2))
                                   for i in range(0, len(b), 8)))
```

Input:

Sender Program

Enter message: Hi

Data encoded and saved

OUTPUT:

Received message: Hi

RESULT:

Thus the Error Correction at Data Link layer is executed